

Assembly Instructions

RadioLab V2.1



Beta Version 0.04
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IMPORTANT

1. Verify the kit contents before you start building!
2. Tools required:
 - a. Basic tools:
 - i. A Phillips screwdriver.
 - ii. Wire cutter
 - iii. A trimmer PPT screwdriver (or a small flathead screwdriver)
 - b. Soldering tools (iron, solder, etc.)
 - c. Multimeter: to check voltages on the board, and also the continuity tests.
 - d. 12V 1A DC power supply
 - e. Speaker 4 or 8 Ohm (to test the audio output)
3. Optional tools:
 - a. Some of the tests outlined require the use of RF Signal Generator, VNA, Oscilloscope, and Spectrum Analyser. If you do not have this equipment then simply ignore these tests as each section will be tested 'On Air' later.

1.0 Kit Contents

The following is a list of the kit contents broken down into each section of the RadioLab.

RadioLab V2.1 - Kit Contents			
Power Section		Prototype Aera	
Power Connector - Solder	1	Standoff 12mm	6
30cm Red and Black wire	1	M3 Screw x 6mm	6
		Lock Washer M3	6
AF Amplifier		M3 Nut	6
Knob	1	Breadboard	1
TDA7266M	1		
Phono Socket 3.5mm	1	Hookup	
Mono Jack Plug 3.5mm	1	Pin to Socket - Jumper Wires (20mm)	4
		Aerial Connector SMA - BNC	1
CPU / DISPLAY		Jumpers	20
Knob	1		
ESP32 Module	1	Enclosure	
Display Module	1	Feet	4
4 Way 0.1" Male	1	M3 x 40mm screw	6
		M3 Spring Washer	10
RF Cables		M3 washer	10
300mm	2	M3 Nut	10
150mm	2	M3 x 15mm Screw	4
		Power Connector - Panel Mount	1
IF Filter			
Crystal 9.2MHz	4	RF Filter	
		Toroid T50-6	2
VFO		Trimmer Capacitor	2
Knob	1	Enamel Wire	1
IF Amplifier		PCB	
Knob	1	Main PCB	1



Remove the 6 screws holding the main PCB from the enclosure and the kit contents are contained in the enclosure.

NB: Keep the M3 Screws / washers / spring washer and nuts safe as they are needed to assemble the unit when the kit is complete.

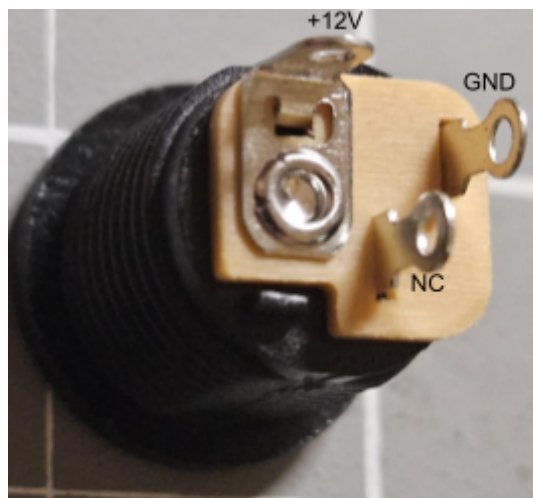
2.0 Assembly Instructions

2.1 Power Section

2.1.1 On the underside of the board there are two pads for power input. Solder wires to GND and +12V pads. Use red wire for +12V and black wire for GND.



2.1.2 Connect +12V (Red Wire) and GND (Black Wire) to the power socket as shown.



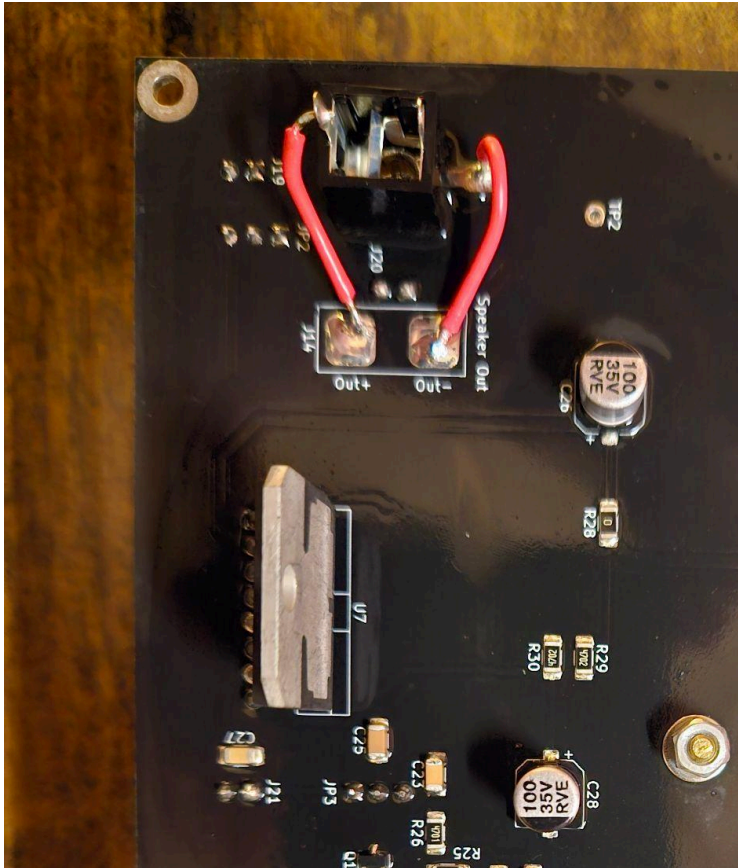
2.1.3 Wire the power socket connector to your +12V Power Supply. Power supply should be linear and be capable of providing 1A (2A when using CW Transmitter Module).



Testing

- Apply power to the board.
- Turn the switch on and the green led should light up.
- Verify the +3v3 and 5V regulators on the 2 RF amplifiers, variable frequency oscillators and display sections of the board.

The Audio Amplifier IC is inserted at the underside of the board and soldered at the top of the board. For protection some silicone rubber or equivalent is recommended to protect these connections from shorting and damaging the amplifier.



Testing

-

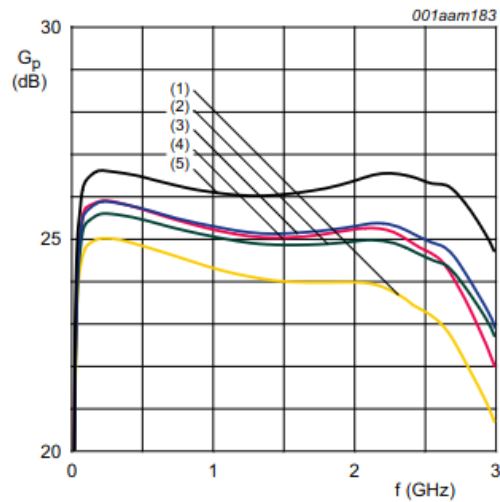
- Use an audio signal generator and apply a 1KHz signal to the Audio Input female header.
- Adjust the volume control (Gain) and the sound should be heard on the speaker.

If you do not have an audio signal generator then connect a stripped wire to the audio input connector and you should hear mains hum on the speaker.

2.3 RF Amplifier Section

Testing

- Use an RF signal generator and apply a 10MHz signal to the Input female header or SMA connector.
- Use an spectrum analyser or oscilloscope to verify the gain of the amplifier.
- See below for a graph of gain vs frequency taken from the datasheet.



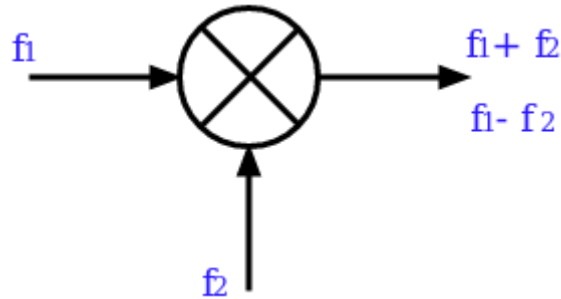
Note : Ignore this testing if you do not have the required equipment.

2.4 Double Balanced Mixer Section



Testing

- Use 2 RF signal generators and apply a 7dBm 10MHz signal to the RF In SMA connector and apply a 7dBm 5MHz signal to the LO In SMA connector.
- Observe the output on a spectrum analyser and check for 5MHz and 15MHz peaks.



Note : *Ignore this testing if you do not have the required equipment.*

2.5 Variable Frequency Oscillator Section

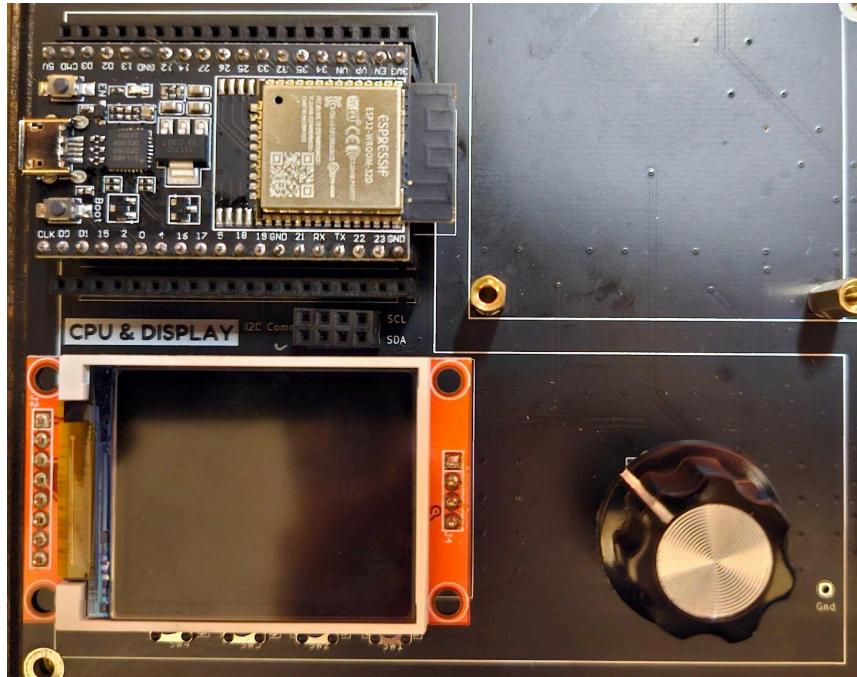


Testing

Testing will be done after the Display section is complete.

2.6 Display and CPU Section

Component placement shown below.



Insert the ESP32 Microcontroller board.

Solder the 4 pin connector to the Display and insert into the board.

Testing

A separate document / video will detail the software installation and testing of this section.

2.7 IF Amplifier Section

Component placement shown below.

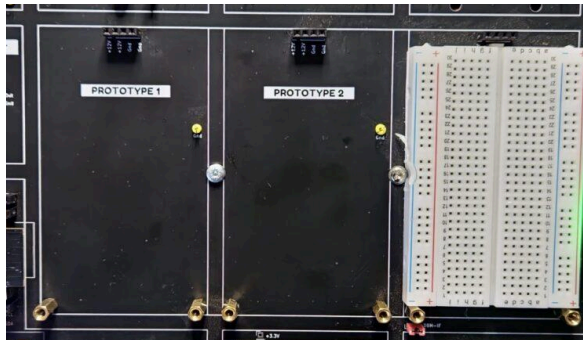


Testing

- Add a jumper at the Pot side of the 3 pin header.
- Use an RF signal generator and apply a 5MHz, -60dBm signal to the Input female header or SMA connector.
- Use an spectrum analyser or oscilloscope to verify the gain of the amplifier.
- The gain of the amplifier should vary approx. 30-60dBm giving an output of -30dBm to 0dBm.

2.8 Prototype Area Section

Component placement shown below.

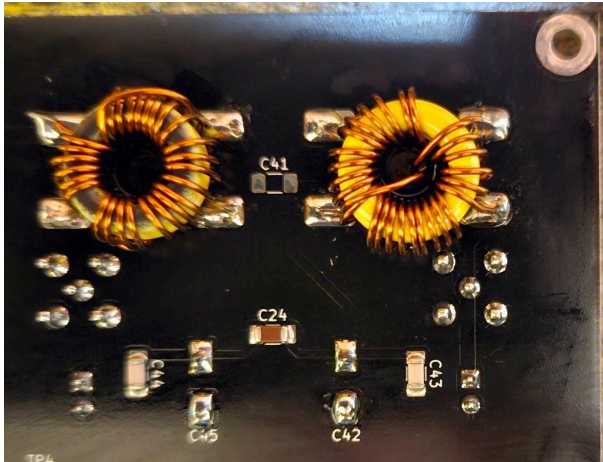


Note :

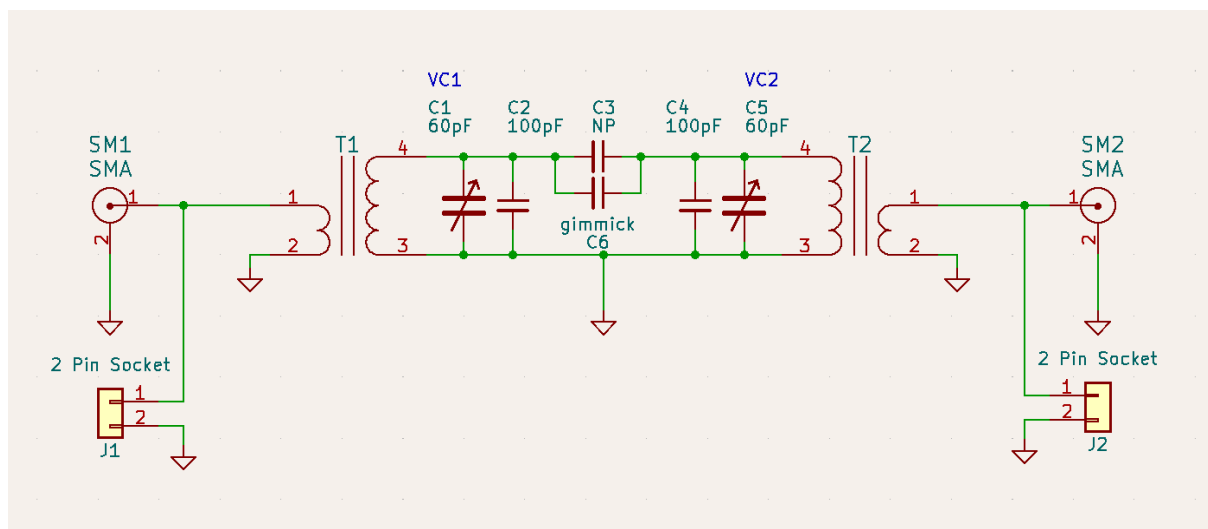
It is important to place the washers and lock washer at the bottom side of the board. This will allow the prototype board to mount flush with the 4 pin female header power connectors.

3.0 RF Filter Assembly Instructions

Component placement shown below.



Schematic shown Below



Coil Winding

- Parts are supplied for the 40M (7MHz) band.
- The primary to secondary for both transformers is 6:21.
- Start off this more than 21 for each secondary and remove turns until you get to the 7MHz range.

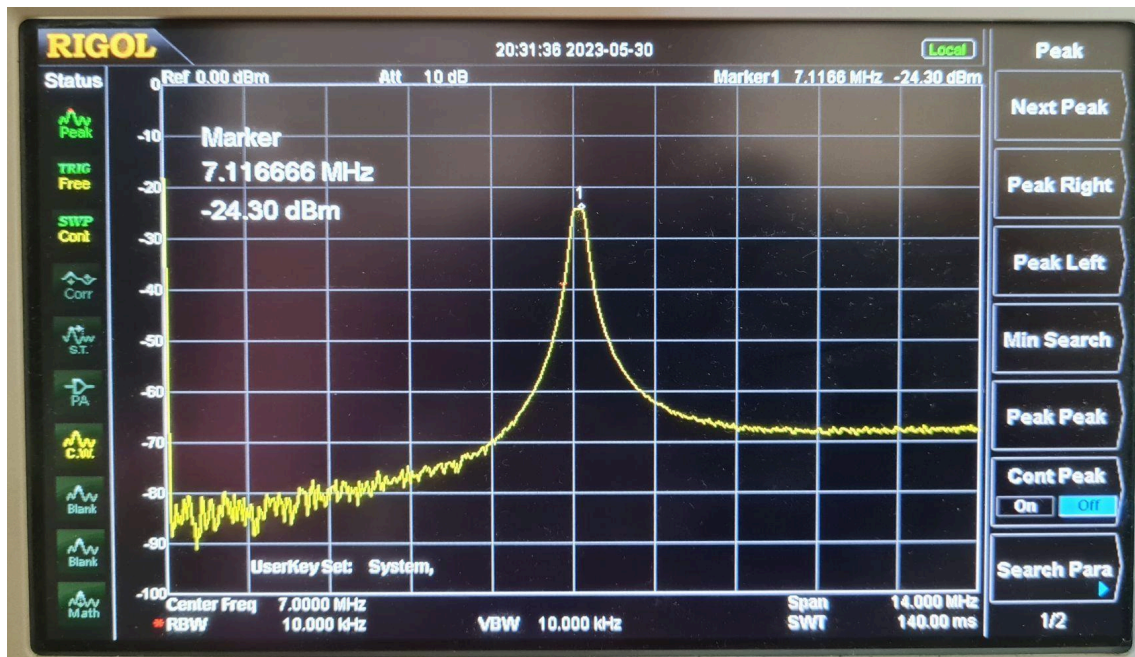
If you don't have experience with RF filters, QRPLabs have a great document explaining the building and testing of filters.

<https://qrp-labs.com/images/bpfkit/bpf3.pdf>

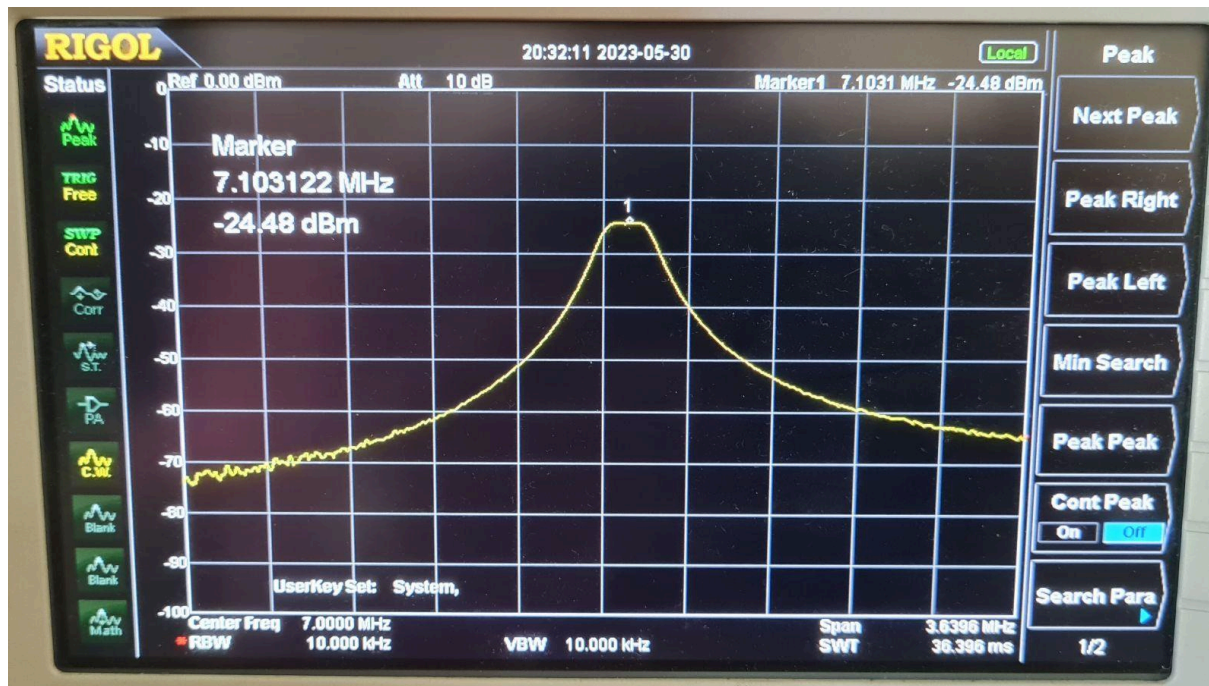
A ceramic screwdriver makes the adjustment easy. Using a metal screwdriver is possible but it will affect the capacitance of the trimmer so it will require adjusting off-peak and removing the screwdriver to see the actual change.



Example of RF Bandpass Filter response.



Example of RF Bandpass Filter response (Zoomed In).



4.0 IF Filter Assembly Instructions

Component placement shown below.



Filter Design

- This filter is primarily designed for SSB.

Testing the crystal filter

- Connect an RF signal generator to the input.
- Connect an oscilloscope to the output.
- Sweep the RF generator around 9.2MHz and observe the signal on the oscilloscope peaking at this frequency.

5.0 System Testing

5.1 Build a Superheterodyne Radio

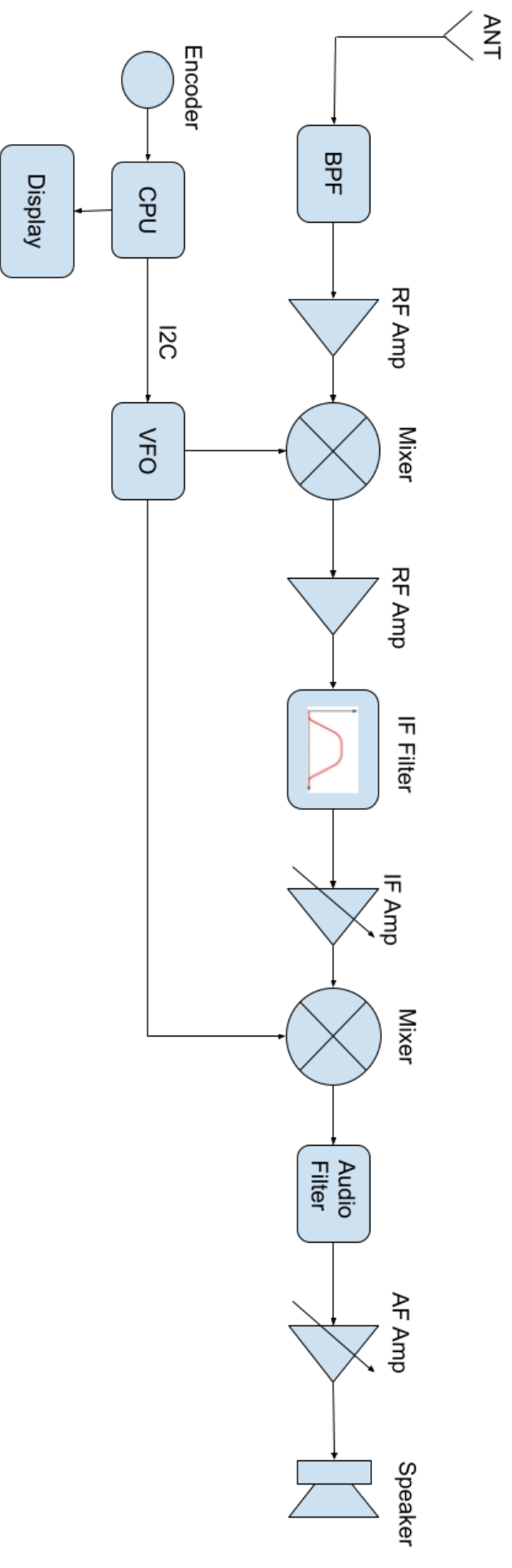
This radio uses all the sections of the radio and therefore is a good way to test the system.

1. Connect Power, 3.5mm Speaker jack and antenna.
2. Insert the jumpers as shown below.

Physical Connections



Superheterodyne Radio Diagram



5.2 Software Setup

Arduino software is available here <https://www.arduino.cc/en/software>
Both IDE v1 and the newer v2 can be used.

To install in Linux <https://docs.arduino.cc/software/ide-v1/tutorials/Linux>

To install in Windows <https://docs.arduino.cc/software/ide-v1/tutorials/Windows>

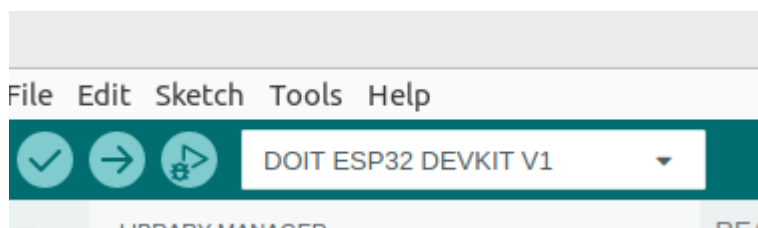
To install in a Mac <https://docs.arduino.cc/software/ide-v1/tutorials/macOS>

5.2.1 Board Setup

Here is a tutorial on setting up the ESP32 board in the Arduino Software.

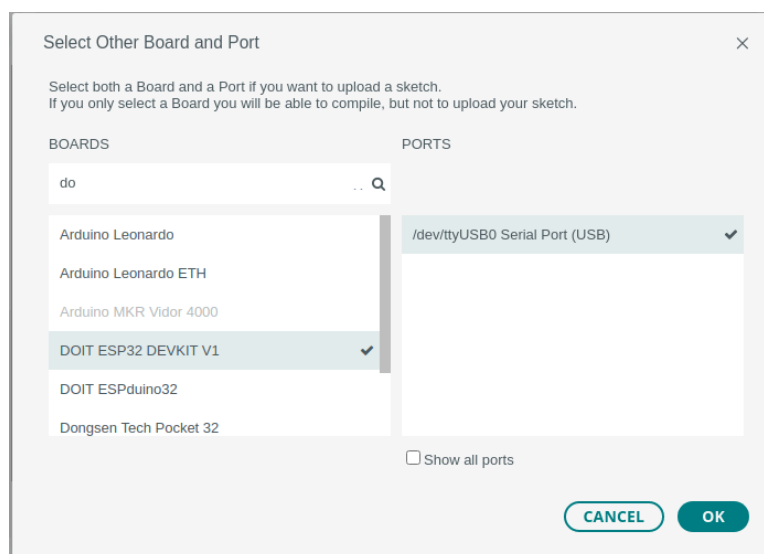
<https://randomnerdtutorials.com/installing-the-esp32-board-in-arduino-ide-windows-instructions/>

The board to select is **DOIT ESP32 DEVKIT V1**



Select the COM port.

In Linux it will be named ttyUSBX and Windows COMX



5.2.2 Arduino Libraries Required

This is a list of libraries that need to be installed using the arduino IDE.

Other libraries are also required. See Software Versions.xlsx for more details.

Ai Esp32 Rotary Encoder by Igor Antoli...
1.7 installed
Easy implement rotary encoder to your application for ESP32 and ESP8266 Supports...
[More info](#)

EasyButton by Evert Arias...
2.0.3 installed
Arduino library for debouncing momentary contact switches, detect press, release, long pre...
[More info](#)
2.0.3 ▼ REMOVE

Etherkit Si5351 by Jason Milldrum...
2.1.4 installed
A full-featured library for the Si5351 series of clock generator ICs from Silicon Labs This libra...
[More info](#)
2.2.0 ▼ UPDATE

MegunoLink by Number Eight Innovation
1.43 installed
Library for storing data in the eeprom, processing serial messages and plotting data...
[More info](#)

PU2CLR SI4735 by
Ricardo Lima Caratti,...

2.1.8 installed

Control your Silicon Labs SI47XX
Family (SI4730, SI4732, SI4731,
SI4734, SI4735, SI4740, SI4745...
[More info](#)

2.1.8 ▼

REMOVE

TFT_eSPI by Bodmer

2.5.31 installed

TFT graphics library for Arduino
processors with performance
optimisation for RP2040,...
[More info](#)

2.5.43 ▼

UPDATE

TFT_eWidget by Bodmer

0.0.5 installed

A TFT GUI widget library A
TFT_eSPI support library
providing button, graph, mete...
[More info](#)

0.0.5 ▼

REMOVE

5.3 Display Library Setup

The TFT_eSPI library needs to be configured for this display. Change the contents of the User_Setup.h file to this.

This file is located under the Arduino libraries path (For example, on macOS, the directory would be ~/Documents/Arduino/libraries/TFT_eSPI. On Windows, it would be My Documents\Arduino\libraries\TFT_eSPI).

User_Setup.h

```
#define USER_SETUP_INFO "User_Setup"
#define ST7735_DRIVER // Define additional parameters below for this display
#define TFT_WIDTH 128
#define TFT_HEIGHT 160
#define ST7735_REDTAB
#define TFT_MISO 15
#define TFT_MOSI 23
#define TFT_SCLK 18
#define TFT_CS 17 // Chip select control pin
#define TFT_DC 2 // Data Command control pin
#define TFT_RST 14 // Reset pin (could connect to RST pin)
#define LOAD_GLCD // Font 1. Original Adafruit 8 pixel font needs ~1820 Bytes in FLASH
#define LOAD_FONT2 // Font 2. Small 16 pixel high font, needs ~3534 Bytes in FLASH, 96
characters
#define LOAD_FONT4 // Font 4. Medium 26 pixel high font, needs ~5848 Bytes in FLASH,
96 characters
#define LOAD_FONT6 // Font 6. Large 48 pixel font, needs ~2666 Bytes in FLASH, only
characters 1234567890:-.apm
#define LOAD_FONT7 // Font 7. 7 segment 48 pixel font, needs ~2438 Bytes in FLASH,
only characters 1234567890:-.
#define LOAD_FONT8 // Font 8. Large 75 pixel font needs ~3256 Bytes in FLASH, only
characters 1234567890:-.
#define LOAD_GFXFF // FreeFonts. Include access to the 48 Adafruit_GFX free fonts FF1
to FF48 and custom fonts
#define SMOOTH_FONT
#define SPI_FREQUENCY 27000000
#define SPI_READ_FREQUENCY 20000000
#define SPI_TOUCH_FREQUENCY 2500000

#define TFT_RGB_ORDER TFT_RGB
```

5.4 Getting the Software

Clone this code.

You can use either the command line git, a GUI git app, or in the simplest method, just download the zip archive of the repo

<https://github.com/Radio-Builder/Version2>

Command Line git

git clone <https://github.com/Radio-Builder/Version2>.git

Build, download and the system should come to life as a 40M receiver:

1. Connect ESP32 to your laptop/PC.
2. Open the "SuperheterodyneRadio.ino" file in the SuperheterodyneRadio directory with the Arduino IDE.
3. Make sure the board and port are selected correctly.
4. Click on the upload button (or click the Upload from the Sketch menu)
5. Wait until Arduino compiles and uploads the code to the board before disconnecting ESP32 from your computer.

Note : For Version 2.1 of the board the Encoder goes backward.

Change

```
//----- Encoder Init -----//  
#define ROTARY_ENCODER_A_PIN 34  
#define ROTARY_ENCODER_B_PIN 35  
to  
//----- Encoder Init -----//  
#define ROTARY_ENCODER_A_PIN 35  
#define ROTARY_ENCODER_B_PIN 34
```